

LESSON 323 (1.5) DUAL, (1.3) INSTRUMENT

Lesson Objective

During this lesson, the student will increase proficiency in instrument flight by review and practice of those procedures listed. In addition, the student will practice emergency and loss of communication procedures.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

Lesson Content (Ground)

Go/no-go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Flight)

Aircraft Checklist Items

Emphasize timeliness to minimize ground time!

- Instrument clearance
 - Instructors, you could either file to pick up your clearance on the ground/in the air OR pick up practice VFR approaches in the air.
 - Make sure your student is NOT the one filing.
 - File KMLB to KMLB and choose an IAF or FAF of your choosing from the approaches in KMLB.
 - In the remarks section, list the approaches desired or “multiple practice approaches.”

- Instrument Cockpit Check
 - Student Tells, Student Does.
 - Have the student read each item from the checklist out loud.
 - Discipline the student to set bugs, bearing pointers, and any auxiliary features.
 - The autopilot check can be conducted now:
 - Engage autopilot.
 - Test roll: left and right.
 - Test pitch: up and down (VS mode).
 - Ensure autopilot can be overridden.
 - Check the red disconnect button.
 - Electric trim test:
 - Check the motion of the trim wheel.
 - Check the S-TEC trim annunciators.
 - Each half-switch should not move the trim wheel.
 - The disconnect switch should stop the wheel.
- Before taxi check:
 - Student Tells, Student Does.
 - Airspeed indicator (PFD and standby): zero.
 - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
 - Standby attitude indicator: standby battery check and adjust the miniature aircraft.
 - Altimeter (PFD and standby): +/- 75 feet of station elevation.
 - Vertical speed indicator (PFD): reads 0 or use current indication.
 - Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
 - HSI: matches compass and turns appropriately.
 - Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.

Departure

- Follow the tower's instructions for departure.
- The student must put the hood or foggles on when safe.
- Climb out:
 - Student Does, Instructor Evaluates.
 - Maintain V_Y .
 - Conduct the 1,000ft check.
 - Level off at a given altitude.
 - Contact departure when handed over by tower. Check in with only heading and altitude initially.

ILS or Localizer

- Tell the student to expect an ILS or localizer approach.
 - Nearby options include the ILS 09R/LOC BC 27L @ KMLB, the ILS 36 @ KTIK, or the ILS 03/21 @ KCOF.
- If picking up the approach request in the air, make sure you show your student how to contact Orlando approach before making them do it.
 - **Weather:**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS)
 - Load the procedure into the GPS. Load each procedure from an IAF transition (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
 - If activating vectors to final, advise your student to set the appropriate approach course.
 - **Radios:**
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - Set the final approach course if required.
 - **Enroute brief**
 - Emphasize the importance of this being a brief and not explaining the plate. Students will tend to miss radio calls as they will be talking over approach. Instructors, make sure you are paying attention to the frequency and also listening to your student briefing the plate. Adjust the comm volumes accordingly.

VOR approach:

- Tell the student to expect a VOR approach starting from a VOR (for example, the VOR 09R full approach at KMLB). Give them some space then tell them to proceed direct to the VOR, then fly outbound.
- Once established on this track, consider engaging the autopilot or pausing the sim to allow the student to run the WIRE checklist.
- Ensure the student completes all the steps of the WIRE checklist:
 - **Weather**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS)
 - Load the procedure into the GPS. Load each procedure from an IAF transition (if ATC does not specify, choose one that makes the most sense) unless ATC states

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the approach will begin from radar vectors.

- Explain the difference between LOAD vs ACTIVATE. Loading the approach preserves waypoints in the flight plan to allow for an activation once ready to commence the approach. Activating the approach deletes all other waypoints and goes direct to whichever transition was selected.

- Radios:

- Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
- Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
- Set the final approach course if required.

- Enroute brief

- Prior to reaching the VOR, point out the G430's prompts to turn early onto the outbound radial so as not to overshoot. Turn to the outbound heading and twist to the outbound radial.
- Emphasize that since an approach clearance has not yet been given, a descent or procedure turn is not authorized. Shortly after, clear them for the approach.
- Ensure the student utilizes proper descent planning when deciding when to execute their procedure turn.
- During the procedure turn, ensure the student sets airspeed to 90 knots, sets flaps to 10°, sets the MFD to the engine page, and sets the inbound course.
- Explain that since there is no glideslope, a proper descent rate can be maintained by setting approximately 1,800RPM (1,500RPM in the simulator) and pitching for 90 knots.
- Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
- Ensure the student levels off at and holds the MDA until the MAP. Remind the student of the 91.175(c) criteria required to land.
- Ensure the student completes the steps for a missed approach correctly:
 - Set full power and pitch up for V_y
 - Upon reaching a positive rate of climb, retract the flaps
 - Turn to the missed approach heading/radial
 - Notify ATC

GPS approach

- Ask ATC to begin a GPS approach from an IAF (for example, NAJBE for the RNAV 09R @ KMLB). If given a direct to fix instruction, proceed to this fix via the PROC page.
- Ensure the student completes all the steps of the **WIRE** checklist:
 - Weather
 - Instruments:
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS).
 - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does

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not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.

- **Radios:**
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - Set the final approach course if required.
- **Enroute brief**
- Point out the G430 prompts to turn just prior to reaching fixes on the approach so as not to overshoot.
 - If autopilot is to be used, engage GPSS mode by pressing NAV twice to ensure this happens automatically.
- Prior to final approach course intercept and when cleared for the approach, ensure airspeed is 90 knots, extend the flaps to 10°, and set the MFD to the engine page.
- At the FAF, remind the student that setting around 1,800RPM and pitching for 90 knots will provide a -500fpm descent rate to follow the glideslope.
- Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.

Circling approach

- On any one of the approaches, request a circle to land.
 - If this cannot be completed at KMLB, it can be completed at X26 without talking to ATC. Make sure to make calls to CTAF and circle at pattern altitude if needed.
- Remind the student of the circling radius airspace.
 - The  symbol indicates expanded circling radius. A lack of this symbol indicates standard circling radius.
- Just prior to the specified distance, the student should go visual without being told, and begin the circle.
 - During their approach briefing, the student should include the direction (left/right) that they will turn when they begin the circle so that they don't have to figure that out now.
- Inform ATC if required.
- Hold the circling MDA throughout the circle.
- Once a landing is assured, descend below the circling MDA, extend the flaps and land.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.

- Preview the next lesson → 324
- Assign Homework:
 - Review approach plates and procedures
 - Review current NOTAMs for the expected approaches

Completion Standards

During the ILS and/or WAAS instrument approach procedures, the student should:

1. Demonstrate accurate localizer and/or course deviation indicator (CDI) interception and tracking and make a transition to the glide slope at the correct point.
2. Maintain the glide slope and localizer/CDI with less than full-scale needle deflection.
3. Maintain an altitude within + 100 feet on the initial and intermediate approach segments.
4. Maintain an altitude that is not more than 100 feet above the MDA.